Transient Couette Flow: Numerical Comparison of Explicit and Implicit Schemes

FTCS and backward-Euler solutions compared with the analytical velocity profile

1. Aim

To compare explicit FTCS and implicit backward-Euler methods for transient viscous momentum diffusion in Couette flow. The computed velocity distributions are assessed against the analytical series solution, with particular attention to the influence of the Fourier number on stability and accuracy.

2. Physical Model and Simulation Parameters

The problem considers fluid initially at rest between two parallel plates. The lower plate remains fixed, while the upper plate is suddenly set into motion at a constant velocity. With no imposed pressure gradient, momentum diffuses through the fluid in the direction normal to the plates.

Parameter	Specification
Kinematic viscosity, nu	0.01
Upper plate velocity, U	1
Final simulation time, t_end	2
Grid points, ny (first case)	81
Explicit Fourier numbers	0.25, 0.45 and 0.55
Implicit Fourier number (target)	5

3. Governing Equation and Boundary Conditions

The governing equation is the one-dimensional unsteady diffusion equation for streamwise velocity, with kinematic viscosity nu. Initially, the fluid velocity is zero throughout the gap. The lower plate has zero velocity, and the upper plate is assigned velocity U at the start of the calculation. A zero pressure gradient is assumed.

4. Numerical Formulations

The explicit solution is obtained using the Forward-Time Central-Space (FTCS) scheme. Its stability condition for this diffusion problem is Fo <= 0.5. The implicit solution uses the backward-Euler formulation, which remains stable at larger Fourier numbers. Both numerical approaches are compared with the analytical series solution at the same physical time.

5. First Grid Case: ny = 81

The first set of calculations uses 81 grid points. Three explicit Fourier numbers (0.25, 0.45 and 0.55) are tested, while the target implicit Fourier number is 5. The supplied results show stable profiles for the two explicit cases below the stability limit and an unstable response when Fo is raised to 0.55.

6. Results for ny = 81

Quantity	Reported result
Kinematic viscosity, nu	0.01
Upper plate velocity, U	1
Grid points, ny	81
Final time	2
Explicit Fourier numbers	0.25, 0.45, 0.55
Implicit Fourier number	5
FTCS stability limit	Fo <= 0.5
Relative L2 error at t = 2, Fo = 0.25	6 x 10^(-4)
Relative L2 error at t = 2, Fo = 0.45	3 x 10^(-4)
Relative L2 error at t = 2, Fo = 0.55	10^15 (unstable)
Relative L2 error at t = 2, implicit Fo = 5	7 x 10^(-3)

The velocity-profile plots show close agreement between the explicit and implicit results and the analytical curve for Fo = 0.25 and 0.45. At Fo = 0.55, the explicit solution develops large oscillations and becomes unbounded. The error-history plot for this case also rises sharply, consistent with violation of the FTCS stability limit.

7. Second Grid Case: ny = 51

A second calculation is presented using 51 grid points. The explicit Fourier-number values remain 0.25, 0.45 and 0.55. The actual implicit Fourier number is reported as approximately 4.73, rather than the target value of 5.0. The source indicates that the explicit solution at Fo = 0.55 is unstable, while the implicit solution remains bounded.

Quantity	Reported result
Kinematic viscosity, nu	0.01
Upper plate velocity, U	1 m/s
Grid points, ny	51
Final time	2 s
Explicit Fourier numbers	0.25, 0.45, 0.55
Implicit Fourier number (actual)	4.73
FTCS stability limit	0.5
Explicit Fo = 0.25 and 0.45 errors	Read from solver output
Explicit Fo = 0.55	Unstable; error grows rapidly
Implicit error at t = 2	Read from solver output

8. Discussion

The comparison demonstrates the role of the Fourier number in the stability of the explicit scheme. For FTCS, values of 0.25 and 0.45 satisfy Fo <= 0.5 and produce bounded velocity profiles. Increasing Fo to 0.55 crosses the

stability threshold and leads to growing numerical oscillations. This criterion is unchanged when the grid is reduced from 81 to 51 points.

The backward-Euler method is stable at the larger Fourier numbers used in the report. However, stability does not by itself guarantee the smallest transient error. The numerical velocity profile should still be compared with the analytical solution at the same physical time. The 51-point grid has a larger grid spacing than the 81-point grid, so its spatial resolution and numerical error may differ.

9. Conclusion

Transient Couette flow was analysed using explicit FTCS and implicit backward-Euler formulations, with the numerical velocity profiles compared against an analytical solution. The study considered two grid resolutions, 81 and 51 points, while varying the Fourier number.

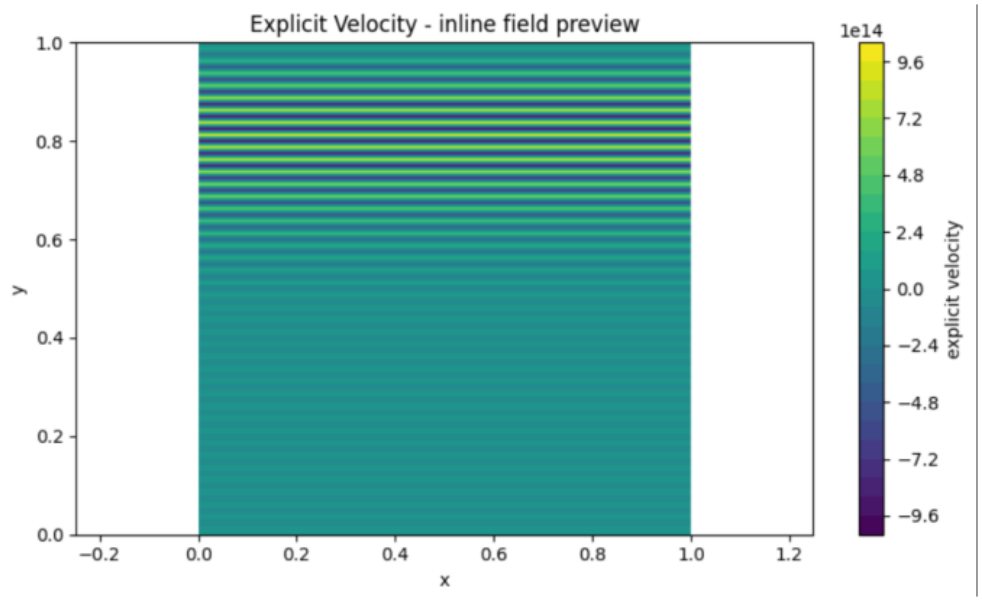
For both grids, the FTCS stability condition is Fo <= 0.5. The explicit cases at Fo = 0.25 and 0.45 remain stable, whereas Fo = 0.55 produces growing oscillations. The implicit method remains bounded at the larger Fourier numbers reported, including an actual value of approximately 4.73 for the 51-point grid.

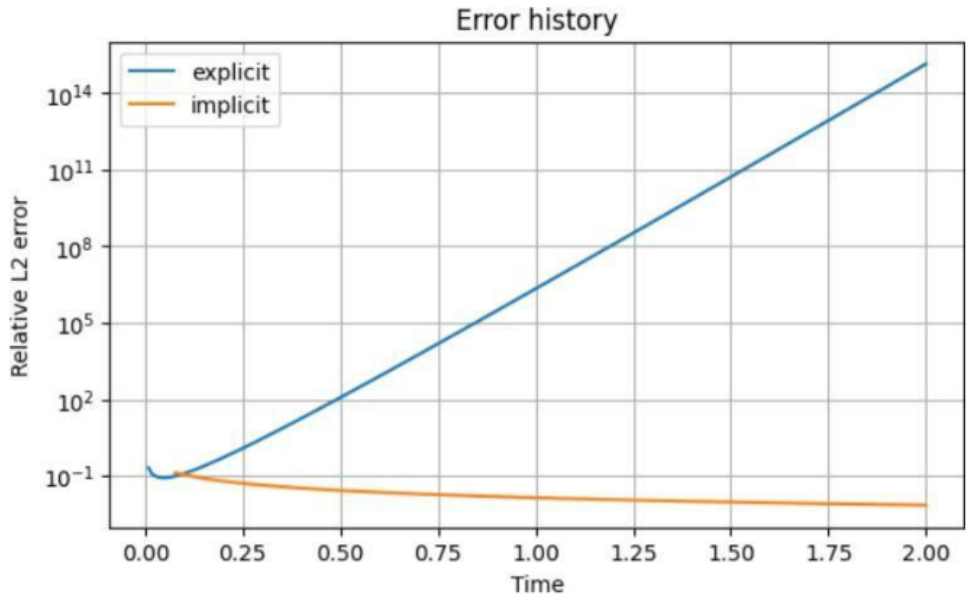
The 81-point case includes reported relative L2 errors, while several error values for the 51-point case are listed as requiring solver output. Therefore, a definitive accuracy comparison between the two grids cannot be made from the supplied data alone.

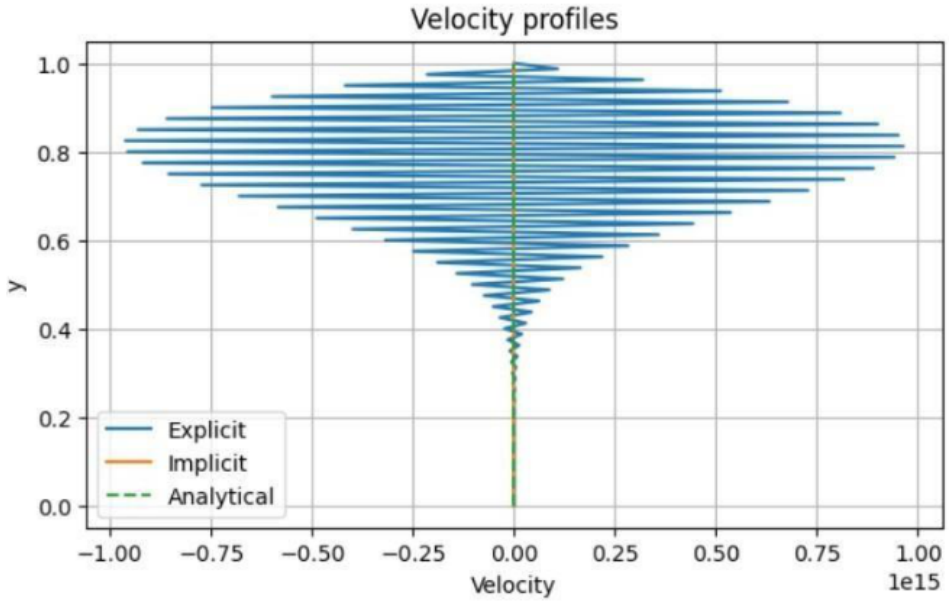
10. Original Simulation Figures

The original velocity profiles, inline velocity fields and error-history plots are retained in the following pages.

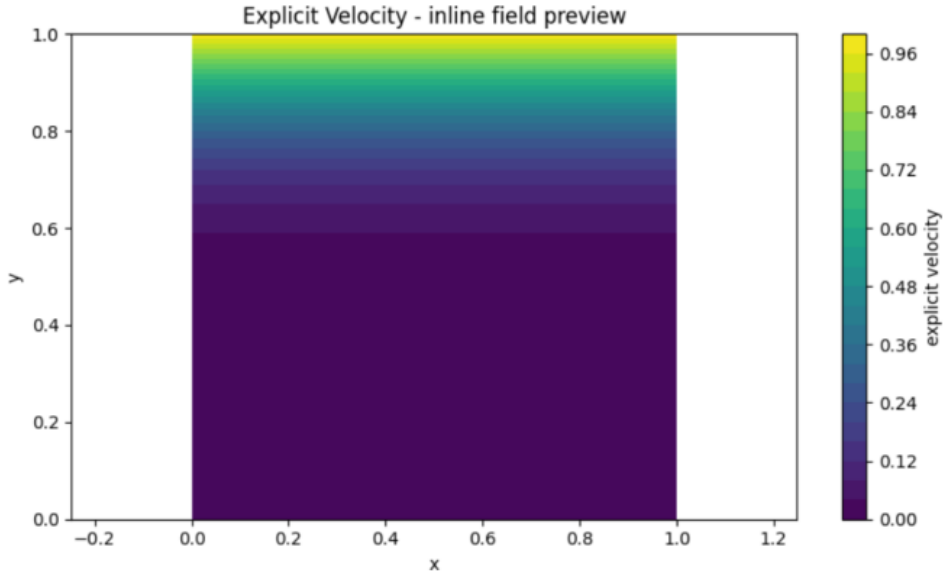
For explicit fourier number 0.55

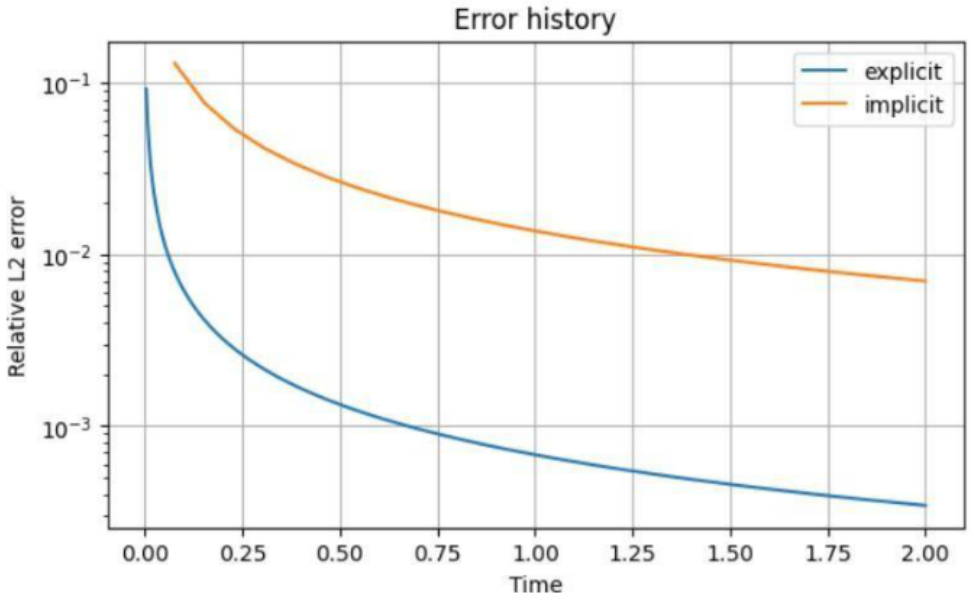


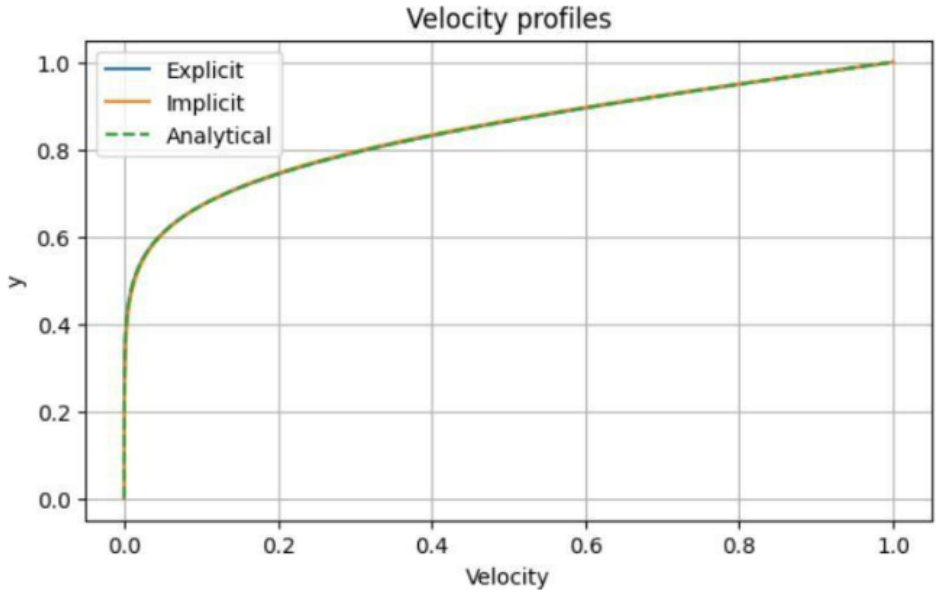




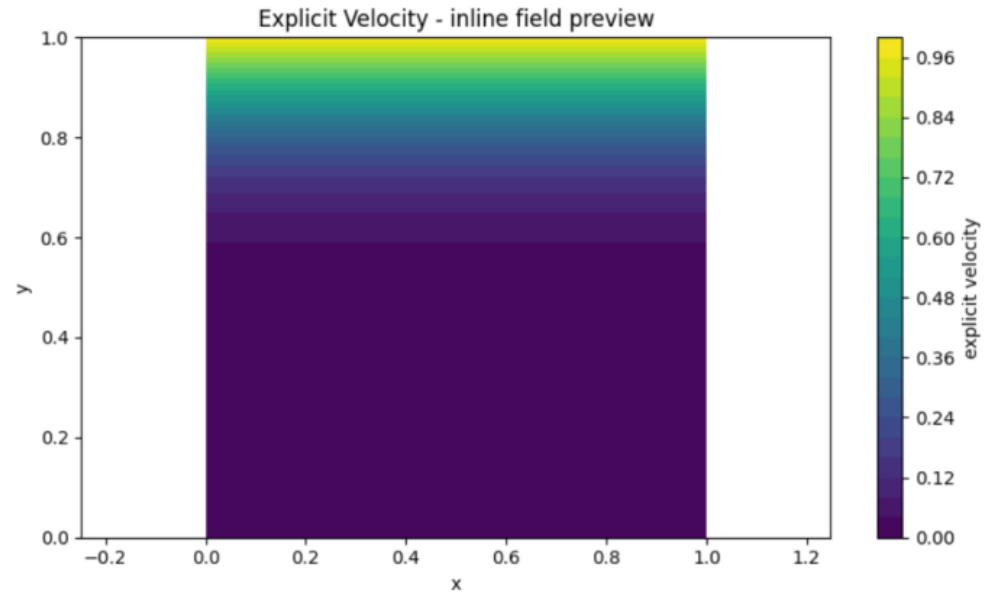
For explicit fourier number 0.45

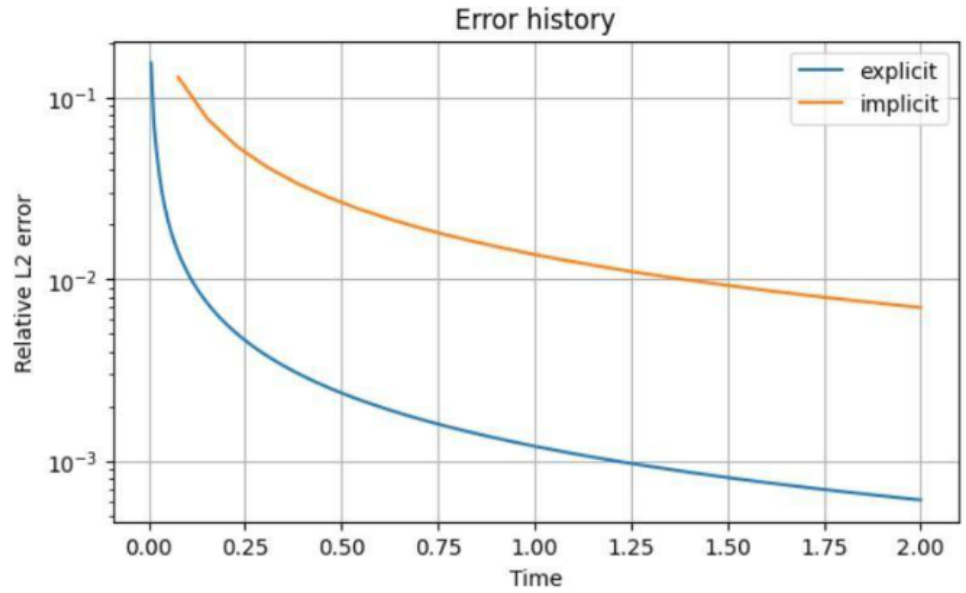


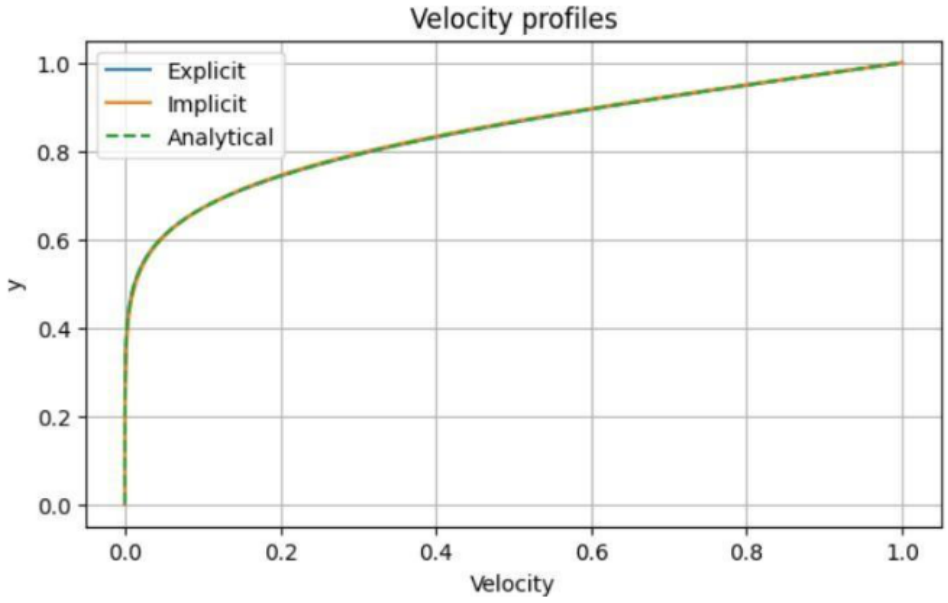

Error history
explicit
implicit
Relative L2 error
10^{-1}
10^{-2}
10^{-3}
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time


Velocity profiles
Explicit
Implicit
Analytical
y
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Velocity

For explicit fourier number 0.25





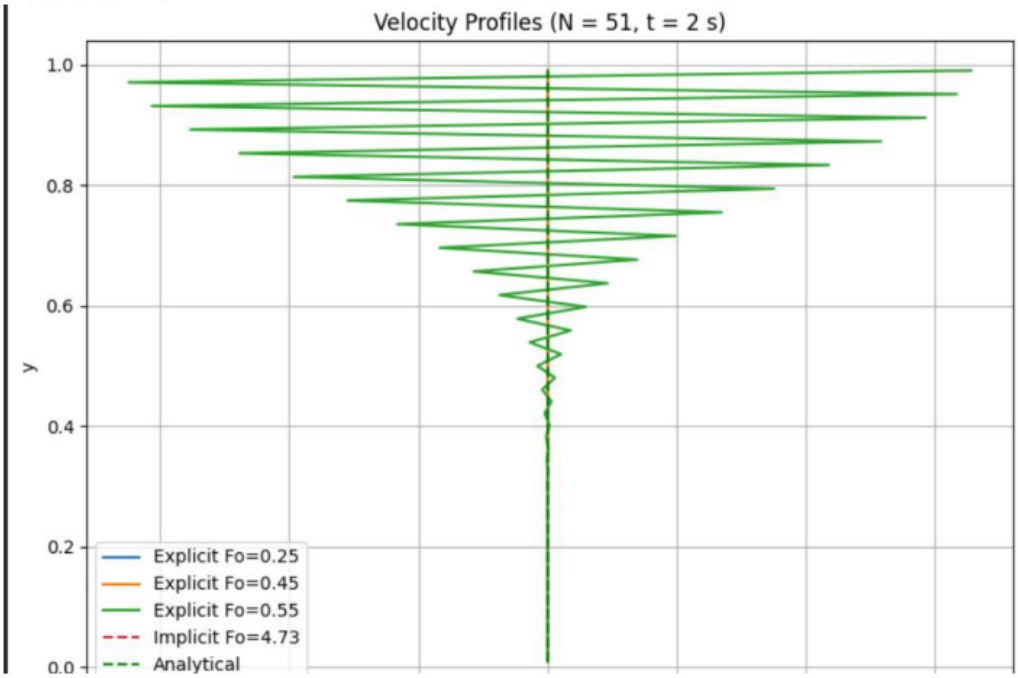


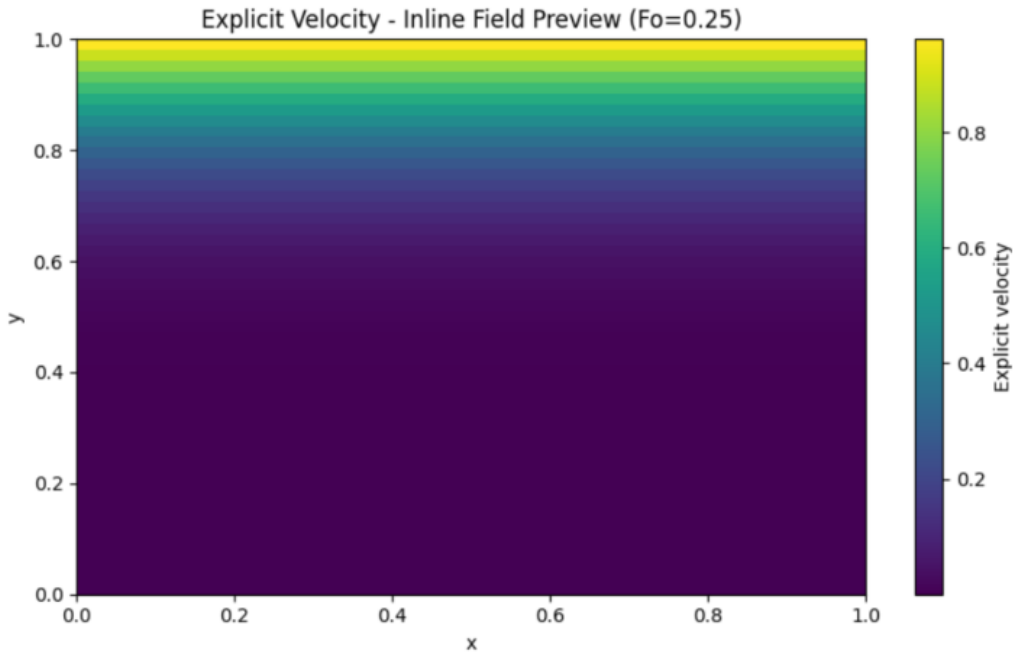
Results:

Parameter	**Result**
Kinematic viscosity, nu	0.01
Upper plate velocity, U	1
Number of grid points, n_y	81
Final simulation time, t_{end}	2
Explicit Fourier number, Fo	0.25, 0.45, 0.55
Implicit Fourier number, Fo	5
Maximum stable Fourier number for FTCS	0.5
Relative L2 error at t=2, explicit Fo=0.25	6×10^{-4}

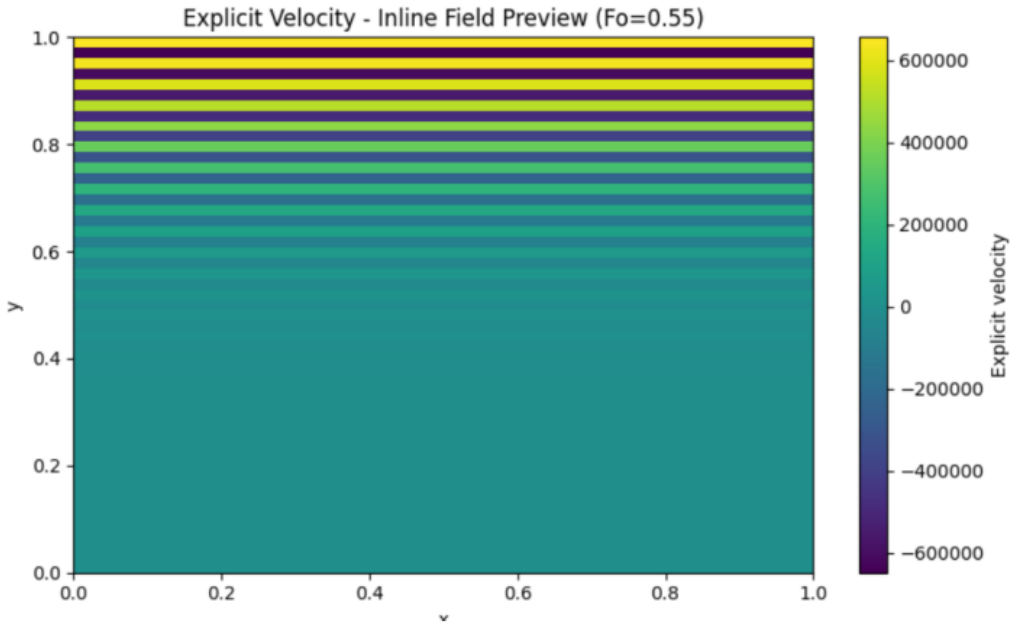
Relative L2 error at t=2, explicit Fo=0.45	$3x10^{-4}$
Relative L2 error at t=2, explicit Fo=0.55	10^{15}
Relative L2 error at t=2, implicit Fo=5	$7x10^{-3}$
Velocity profile for Fo=0.25 and 0.45	Closely matches analytical solution
Velocity profile for Fo=0.55	Unstable, with unbounded oscillations
Overall stability	Explicit stable for Fo\leq0.5\); unstable for Fo>0.5

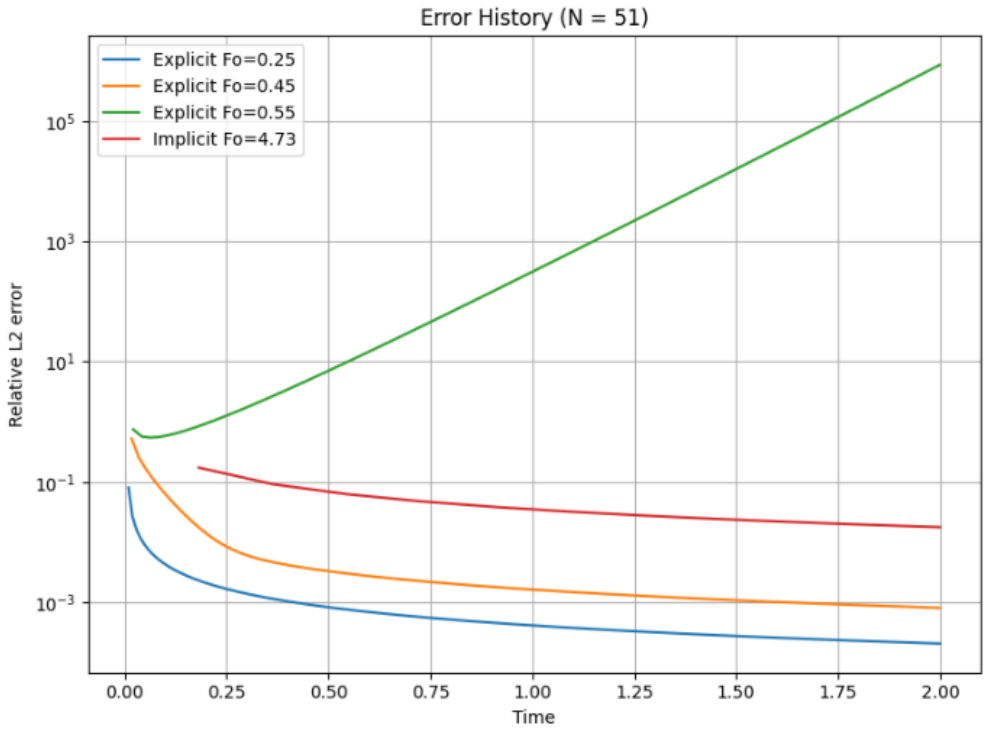
Results at N=51





Explicit Velocity - Inline Field Preview (Fo=0.45)






```
RESULTS FOR N = 51
----------------------------------------------------------------
Grid cells: 51
Grid spacing: 0.0196078431372549
Viscosity: 0.01
Final time: 2.0
Implicit actual Fo: 4.729090909090909
Implicit final relative L2 error: 1.770339e-02
```

Results (N = 51)

Parameter	Result
Kinematic viscosity, (\nu)	0.01
Upper plate velocity, (U)	1 m/s
Number of grid points, (n_y)	51
Final simulation time, (t_{\mathrm{end}})	2 s
Explicit Fourier number, (Fo)	0.25, 0.45, 0.55
Implicit Fourier number, (Fo)	4.73 (actual)
Maximum stable Fourier number for FTCS	0.5
Relative L2 error at (t=2), explicit (Fo=0.25)	Read from solver output
Relative L2 error at (t=2), explicit (Fo=0.45)	Read from solver output
Relative L2 error at (t=2), explicit (Fo=0.55)	Unstable; error grows rapidly

Relative L2 error at (t=2), implicit	Read from solver output
Velocity profile for (Fo=0.25) and (Fo=0.45)	Stable profiles; compare with analytical solution
Velocity profile for (Fo=0.55)	Unstable, with growing oscillations
Overall stability	Explicit stable for (Fo \leq 0.5); unstable for (Fo > 0.5)

Conclusion:

The transient Couette flow problem was investigated using the explicit Forward-Time Central-Space (FTCS) method and the implicit backward-Euler method. The numerical solutions were compared with the analytical solution for viscous momentum diffusion between two parallel plates. The upper plate velocity was maintained at 1 m/s, the lower plate remained stationary, the kinematic viscosity was 0.01 m^2/s, and the final simulation time was 2 s.

Comparison of grid resolution: The original simulation used 81 grid points, whereas the additional simulation used 51 grid points. Reducing the number of grid points increased the grid spacing and changed the time step required to achieve the selected Fourier number. Therefore, the numerical error and resolution of the transient velocity profile may differ between the two grids.

Effect of Fourier number on explicit-scheme stability: In both grid cases, the explicit FTCS method has a theoretical stability requirement of $Fo \leq 0.5$ for the one-dimensional diffusion equation. The cases with $Fo = 0.25$ and $Fo = 0.45$ are within this stability limit, while $Fo = 0.55$ exceeds it and is expected to produce growing numerical oscillations. This stability criterion remains unchanged when the grid is changed from 81 to 51 points.

Comparison with the implicit method: The backward-Euler method remains stable for larger Fourier numbers. For the $N = 51$ simulation, the actual implicit Fourier number was approximately 4.73, compared with the target value of 5.0. The implicit solution remained bounded in the plotted results. However, stability alone does not guarantee the smallest transient error, so the numerical solution should be compared with the analytical profile at the same physical time.

Overall conclusion: The comparison demonstrates that both grid resolution and Fourier number influence the numerical results. The 81-point grid provides finer spatial resolution than the 51-point grid, while the explicit stability limit remains $Fo \leq 0.5$ for both cases. The explicit method becomes unstable when $Fo = 0.55$, whereas the implicit method can operate with a larger Fourier number. A quantitative conclusion about which grid is more accurate requires the actual relative L2 errors for both grids at $t = 2$ s.